# Power BI Dashboard Analysis

## Global Airbnb Market Intelligence — Dashboard Storyline

This dashboard is built to answer one overarching question, broken into three progressively deeper layers:

**"Where should we focus attention in the global short-term rental market — and why?"**

**Page 1 (Overview)** answers: Which markets are performing best, at a glance?

**Page 2 (Market Deep Dive)** answers: Why — what's driving those differences underneath the surface?

**Page 3 (Host & Performance)** answers: Who is actually winning within those markets, and does conventional wisdom about "good hosting" hold up?

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### Page 1 → Page 2: from "which city" to "why that city"

The Overview page establishes the headline finding: **Paris leads on both occupancy (61.25%) and average price ($321.23)** — an unusual combination, since higher prices normally suppress demand. **Barcelona lags on both dimensions (41.05% occupancy, $240.62 average price)** — equally unusual, since lower prices would normally stimulate demand.

This is the hook. A city-level number can tell you that something interesting is happening, but not why. That question — why does Paris behave this way, and why does Barcelona — is what pulls the viewer into Page 2.

**Page 2 supplies the answer, in three parts:**

1. **Composition:** Based on **room type**, the Airbnb market is heavily dominated by Entire **Home/Apt listings (73.0%)**, followed **by Private Rooms (25.8%)**, while **Hotel Rooms and Shared Rooms together account for less than 2%** of the total inventory. **Entire Home/Apt and Private Room listings also tend to command higher average prices** than the other room types. Cities with a higher concentration of these in-demand room types, particularly when combined with popular residential property types, tend to exhibit **stronger occupancy and listing demand**, which can translate into higher revenue potential.
   This pattern is particularly evident in **Paris and London, which contribute a larger share of overall revenue compared with the other cities**. Their stronger revenue performance can be associated with a favorable combination of **high-demand room types (Entire Home/Apt and Private Room)** and suitable property types, resulting in stronger occupancy and revenue generation. **However, this should be interpreted as an association rather than a direct causal relationship**, since revenue is also influenced by factors such as pricing, location, seasonality, and listing availability.
2. **Pricing & Occupancy**: There is **no consistent relationship between price and occupancy across neighborhoods.** Within the **mainstream price range of approximately $100–$400 per night**, occupancy varies considerably even among similarly priced listings. This suggests that factors such as **location, property quality, neighborhood appeal, and type of property** may have a stronger influence on demand than price alone.

The analysis also highlights **a small luxury segment (>$1,000 per night)** where a limited number of high-priced listings still achieve **moderate occupancy**, indicating a different customer segment with greater willingness to pay. Overall, the market appears to serve **two distinct segments**: a large mainstream market with most listings priced between **$100–$400/night**, and a smaller luxury market where customers are less price-sensitive. Therefore, **pricing strategy should be considered alongside location, property characteristics, and target customer segment rather than occupancy being explained by price alone.**

3. **Granularity**: The analysis covers around **216,000 individual Airbnb listings**, allowing us to compare properties based on **city, neighborhood, room type, property type, and host characteristics**. The **78 million daily calendar records** are then summarized at the **listing–month level** to calculate metrics such as **occupancy, availability, and estimated revenue**. This level of detail helps identify **which types of properties and locations have higher demand, occupancy, pricing, and revenue**, while making the data easier to analyze than working with daily records.

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## Page 2 → Page 3: from “the market” to “the host”

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Page 2 established that occupancy varies enormously even among similarly-priced neighbourhoods. That naturally raises the next question: **is the variation coming from the host, not the listing or the location?**

Page 3 tests two of the most obvious hypotheses a stakeholder would raise:

**Hypothesis 1:** “**Superhosts must perform better — that’s the whole point of the badge.”** The data says otherwise: Superhosts show **lower** occupancy than non-superhosts (52.24% vs. 56.83%), at essentially identical price points. The Superhost badge must be reflecting service quality metrics (response rate, cancellation rate, review scores) — not demand generation. This is a genuinely useful finding precisely because it contradicts the intuitive assumption a business stakeholder would likely walk in with.

**Hypothesis 2: “More experienced hosts must perform better — practice makes perfect.”** Partially true, but not linearly: occupancy climbs steadily from 46% (0-2 years) to a peak of 65.57% (6 10 years) — then **dips slightly** for 10+ year veterans, whose pricing also ticks back up. This suggests a “sweet spot” rather than a straight line — established-but-still-hungry hosts (6-10 years) may be the market’s most effective operators, while the most veteran hosts may shift toward fewer, higher-value listings rather than maximizing fill rate.

**Both findings converge on the same underlying insight the whole dashboard has been building toward:** performance in this market is not explained by the “obvious” signals — status badges, tenure, or price alone. It’s explained by something more granular that this dataset can gesture at but not fully resolve — likely a combination of listing quality, exact micro-location, and operator strategy that would require a follow-up analysis (e.g., text-mining listing descriptions, amenities data, or review sentiment) to fully unpack.

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### The final, unified business insight

If you had to distill the entire three-page investigation into a single takeaway for a stakeholder:

> ***Headline performance differences between cities are real, but they are driven by market composition and neighbourhood-level variation, not by simple, portable rules like "charge more" or "hire a Superhost." Paris outperforms not because of some uniform city-wide advantage, but likely because of a richer mix of property types and neighbourhoods. Within any single market, the conventional signals of quality — Superhost status, host tenure, even price itself — are weak or even counterintuitive predictors of occupancy. Any pricing or investment strategy built on these surface-level signals alone would likely underperform; the real drivers of performance sit at the neighbourhood and listing level, and would require deeper investigation (e.g., amenities, review content, exact location) to fully explain. What this means practically, if this were a real business context: For market entry/investment: prioritize neighbourhood-level due diligence over city-level averages — the 20x within-city price spread means "Barcelona" or "Paris" as a label tells you very little on its own. For pricing strategy: don't assume lowering price will reliably increase occupancy (the scatter plot shows no clean relationship) — and don't assume higher price signals a failing listing either. For host support/onboarding programs: don't assume Superhost promotion campaigns will move occupancy — if anything, the data suggests Superhost and occupancy optimization may be two different goals requiring different strategies. For further research: the next logical step beyond this dataset would be listing-level features — amenities, photos, review text, response time — since city, price, host tenure, and badge status collectively explain surprisingly little of the performance variation observed.***